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Time and motion in physics: the Reciprocity Principle, relativistic invariance of the lengths of rulers and time dilatation

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Abstract

Ponderable objects moving in free space according to Newton's First Law constitute both rulers and clocks when one such object is viewed from the rest frame of another. Together with the Reciprocity Principle this is used to demonstrate, in both Galilean and special relativity, the invariance of the measured length of a ruler in motion. The different times: 'proper', 'improper' and 'apparent' appearing in different formulations of the relativistic time dilatation relation are discussed and exemplified by experimental applications. A non-intuitive 'length expansion' effect predicted by the Reciprocity Principle as a necessary consequence of time dilatation is pointed out.

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1 Introduction

The standard text-book presentation of special relativity follows closely that of Einstein's seminal paper of 1905 [1] in basing the theory on the Special Relativity Principle, classical electromagnetism and the postulate of constant light speed. However an alternative and conceptually simpler approach to the physics of space and time, in the absence of gravitational fields, is possible in which it is not necessary to consider light signals, classical electromagnetism, or indeed, any dynamical theory whatsoever. The Lorentz transformation (LT) was first derived in this way by Ignatowsky [2] in 1910. Purely mathematical considerations imply, in such a derivation of the LT, the existence of a maximum relative velocity, V, of two inertial frames. Use of relativistic kinematics then shows that V is equal to the speed of light, c, when light in identified as a manifestation of the propgation in space-time of massless particles –photons [3]. In this way Einstein's mysterious second postulate is derived from first principles. The fundamental axiom underlying such an approach is the Reciprocity Principle (RP) [4, 3], discussed in Section 3 below, relating the the relative velocities of two inertial frames. Derivations of the LT and the parallel velocity addition formula based on the RP and other simple axioms are given in Ref. [3].

In the present paper the space-time properties of ponderable[1] physical bodies in free space, as described by Newton's First Law of mechanics, are used together with the RP, to demonstrate the invariance of the measured length of a ruler in uniform motion. The proof given is valid in both Galilean and special relativity, since Newton's First Law and the RP hold in both theories.

The analysis presented is based on a careful definition of physical time concepts. In particular, the 'frame time' or 'proper times' that appear in in the RP, are distinguished from the 'improper time' or 'apparent time' (of a moving clock) that appear in the Time Dilatation (TD) relation of special relativity.

The paper is organised as follows: The following section contains an elementary discussion of the concepts of 'space', 'time' and 'motion' in physics, in connection with Newton's First Law. In Section 3, the RP is introduced and discussed in relation to Newton's First Law. It is pointed out that, because of the RP, 'rulers are clocks' and 'clocks are rulers' when the motion of ponderable bodies in free space is considered. In Section 4 the RP is used to demonstrate the invariance of the measured length of a uniformly moving ruler. In Section 5 the operational meanings of the time symbols appearing in the TD formula of special relativity are discussed. This may be done in a 'clock oriented' manner in terms of 'proper' and 'improper' times of the observed clock, or in an 'observer oriented' manner in terms of the proper time of the observer's local clock and the 'apparent time', as seen by the observer, of the moving clock. Two specific experiments are described to exemplify the operational meanings of the time symbols in the TD formula. A non-intuitive 'length expansion' effect is found to relate similarly defined spatial intervals corresponding to the observation of an event either in the rest frame of the clock, or in a frame in which it is in uniform motion.

[1] That is bodies, with a non-vanishing Newtonian mass, which may be associated with an inertial frame in which the body is at rest. No such frame may be associated with a massless object.

The results of the present paper show that the 'length contraction' effect and the correlated 'relativity of simultaneity' effect of conventional special relativity do not exist. A detailed discussion of the reason for the spurious nature of these effects of conventional special relativity theory may be found in Refs. [5, 6, 7, 8, 9, 10] .

However, a genuine 'relativistic length contraction' effect does occur when distances between spatial coincidences of moving objects are observed from different inertial frames [11]. Also a genuine 'relativity of simultaneity' effect occurs when clocks at rest in two different inertial frames are viewed from a third one [12, 13]. An alternative derivation, directly from the RP, of the invariance of the measured spatial separation of two objects at rest in the same inertial frame as well as the absence of the conventional 'relativity of simultaneity' effect is given in Ref. [9].

2 Physical time and Newton's First Law of Mechanics

In physics the concepts of 'time' and 'motion' are inseparable. In a world in which motion did not exist the physical concept of time would be meaningless. Similarly the physical concepts of 'space' and 'motion' are inseperable. Without the concept of space, no operational definition of motion is possible. The concept of historical time –the time of the everyday world of human existence– requires the introduction of the further, and equivalent, concepts of 'uniform motion' and 'cyclic motion with constant period'. For example, the unit of time the 'year' is identified with the (assumed constant) period of rotation of the Earth about the Sun.

The idea of uniform motion entered into physics in a quantitative way with the formulation of Newton's First Law [14]

`Every body continues in its state of rest, or uniform motion in a right line unless it is compelled to change that state by forces impressed upon it.`

This law gives an operational meaning to the physical concept of 'uniform motion'. It is defined by observations of the position of any ponderable object in 'free space' i.e. in the absence of any mechanical interaction of the object with other objects. There is a one-to-one correspondence between such a ponderable object and an 'inertial frame' of relativity theory. As will be discussed in the following section, one such ponderable object, O, constitutes both a ruler and a clock for an observer in the rest frame of another such object, O', and *vice versa*.

When time is measured by using a cyclic physical phenomenon, e.g. an analogue clock, time measurement reduces to recording the result of a spatial (or angular) measurement. There is a one-to-one correspondence between the spatial coincidence of a stationary 'mark' on the face of the clock and a moving 'pointer', constituted by the hand of the clock, and the time measurement [6]. A 'time interval' is measured by the angular separation of two such 'pointer-mark coincidences'. The implicit assumption is that the motion of the

pointer is 'uniform'. There is an evident logical circularity here since 'equal' time intervals measured by such an analogue clock assume that the angular velocity of the hand is constant, whereas constant angular velocity is established by observation of equal angular increments for equal time time intervals (i.e. also equal angular increments) recorded by a second clock of supposedly known uniform rate. In practice, this conundrum is resolved by an appeal to physics. For example, an undamped pendulum in a uniform gravitational field is predicted, by the laws of mechanics, to have a constant period of oscillation. Quantum mechanics predicts the same transition frequency and mean lifetimes for two identical atoms in the same excited state, in the same physical environment, etc.

Measurements of 'time' are then ultimately observations of spatial phenomena, e.g. the time measurement corresponding to observation of the number displayed by a digital clock is a spatial perception. This will also be the case for time measurements related to observation of two ponderable objects O and O' in motion in free space that will now be discussed.

3 The Reciprocity Principle: rulers are clocks, and clocks are rulers

Consider two non-interacting ponderable objects O and O', with arbitary motions in free space. They are placed at the origins of inertial coordinate systems S and S' with axes orientated so that the x and x' axes are parallel to the relative velocity vector of O and O'. Without any loss of generality for the following discussion, it may be assumed that O and O' lie on the common x-x' axis.

The Reciprocity Principle (RP) [4, 3, 9] is defined by the equation:

$$v = v_{\mathrm{O'O}} = \left.\frac{\partial x_{\mathrm{O'O}}}{\partial t}\right|_{x'_{\mathrm{O'}}x_{\mathrm{O}}} = -\left.\frac{\partial x'_{\mathrm{OO'}}}{\partial t'}\right|_{x_{\mathrm{O}}x'_{\mathrm{O'}}} = -v_{\mathrm{OO'}} \tag{3.1}$$

where $x_{\mathrm{O'O}} \equiv x_{\mathrm{O'}} - x_{\mathrm{O}}$ and $x'_{\mathrm{OO'}} \equiv x'_{\mathrm{O}} - x'_{\mathrm{O'}}$, or in words: 'If the velocity of O' relative to O is $\vec{v}$, the velocity of O relative to O' is - $\vec{v}$'. In many discussions of special relativity, the RP is taken as 'obvious' and is often not even declared as a separate axiom. This is the case, for example, in Einstein's 1905 special relativity paper [1]. However, as first demonstrated by Ignatowsky in 1910 [2], it is sufficient, together with some other weaker axioms such as the homogeneity of space or single-valuedness of the transfomation equations, to derive [3] the space-time Lorentz transformation and hence the whole of special relativity theory.

Eqn(3.1) looks very similar to the equation defining the relative velocity of two objects A and B as observed in a *single* inertial reference frame (say S):

$$v_{\mathrm{AB}} \equiv v_{\mathrm{A}} - v_{\mathrm{B}} = \frac{d(x_{\mathrm{A}} - x_{\mathrm{B}})}{dt} \equiv \frac{dx_{\mathrm{AB}}}{dt} = -\frac{dx_{\mathrm{BA}}}{dt} = -v_{\mathrm{BA}} \tag{3.2}$$

The crucial difference is the appearence in the RP, (3.1), of *two different times* t and t'. The time t is the 'frame time' of S. i.e. the time registered by a synchronised clock at rest, at any position in S, according to an observer also a rest in S. The frame time t' is similarly

defined by an array of synchronised clocks at rest in S'. Eqn(3.1) (and its integral) gives a relation between the times t and t' Both t and t' correspond to 'proper times' of clocks at rest, whereas, as explained in Section 4 below, the Lorentz transformation relates instead a proper time to an 'improper time' –the observed time of a clock in uniform motion.

Suppose now that O and O' are equipped with local clocks that are observed to run at exactly the same rate when they are both at rest in the same inertial frame. The direction of the relative velocity vector $\vec{v}$ of O' relative to O is such that they are approaching each other at the frame times t and t'. The spatial separations of O and O' in S and S' are $\ell(t)$ and $\ell'(t')$ respectively, at times t and t'. Using the RP, a spatial coincidence of O and O' will be observed at the time

$$t_{\mathrm{OO'}} = t + \frac{\ell(t)}{v} \tag{3.3}$$

in S, and

$$t'_{\mathrm{OO'}} = t' + \frac{\ell'(t')}{v} \tag{3.4}$$

in S'. The OO' coincidence event will be *mutually simultaneous* in the frames S and S'.

Note that the OO' spatial coincidence that is mutually simultaneous in S and S' constitutes a pair of *reciprocal* pointer mark coincidences. In S the mark is at the position of O and the moving pointer at the position of O', whereas in S' the position of O' constitutes the mark and the position of O the pointer. A corollary is that all such pairs of reciprocal pointer mark coincidences are mutually simultaneous. This is the basis of the 'system external synchronistation' [15] as introduced in Einstein's first special relativity paper [1] to synchronise clocks at rest in different inertial frames when they are in spatial coincidence.

The observation of the OO' coincidence event in both frames can be used to give a condition that *any* other pair of events, one observed in S, the other observed in S' are mutually simultaneous. If the time of an event in S is $\tilde{t}$ and another event in S' is $\tilde{t}'$ they will be 'mutually simultaneous' providing that:

$$\tilde{t}' - \tilde{t} = t'_{\mathrm{OO'}} - t_{\mathrm{OO'}} \tag{3.5}$$

Combining (3.3)-(3.5) gives:

$$\tilde{t}' - \tilde{t} = t'_{\mathrm{OO'}} - t_{\mathrm{OO'}} = t' - t + \frac{\ell'(t') - \ell(t)}{v} \tag{3.6}$$

If now events occuring at times t in S and t' in S' are mutually simultaneous, it follows from (3.5) and (3.6) that $\ell(t) = \ell'(t')$, so that *events which occur when O and O' have the same spatial separation in S and S' are mutually simultaneous.* A special case occurs if the clock arrays in S and S' are *mutually synchronised* so that $\ell(t) = \ell'(t' = t)$. There is then a direct correlation between either t or t' and the spatial separation of O and O': *When mutually synchronised clocks in the frames S and S' have the same reading, O and O' have the equal spatial separations in S and S'*, and conversely, *When O and O' have equal spatial separations in the frames S and S', mutually synchronised clocks in S and S' have the same reading.*

The dependence of ℓ on t in Eqn(3.3) and ℓ' on t' in Eqn(3.4) means that each of the objects may be considered to be an 'inertial clock' by an observer in the rest frame

of the other one. That is, t is measured by the spatial separation of O' from O in S and t' is measured by the spatial separation of O from O' in S'. Conversely, after mutual synchronisation of the clock arrays in S and S' at the instant when O and O' are in spatial coincidence, t measures the spatial separation of O' and O in S (and so is effectively a ruler in this frame) while t' measures the spatial separation of O' and O in S', constituting a ruler in this frame. Matching of these measurements of the separation of O and O' with the lengths of physical rulers at rest in S and S' is now used to demonstrate the invariance of the measured length of the length of a ruler in uniform motion –that is, the absence of any relativistic length contraction effect– in this case.

4 Invariance of the measured length of a ruler in uniform motion

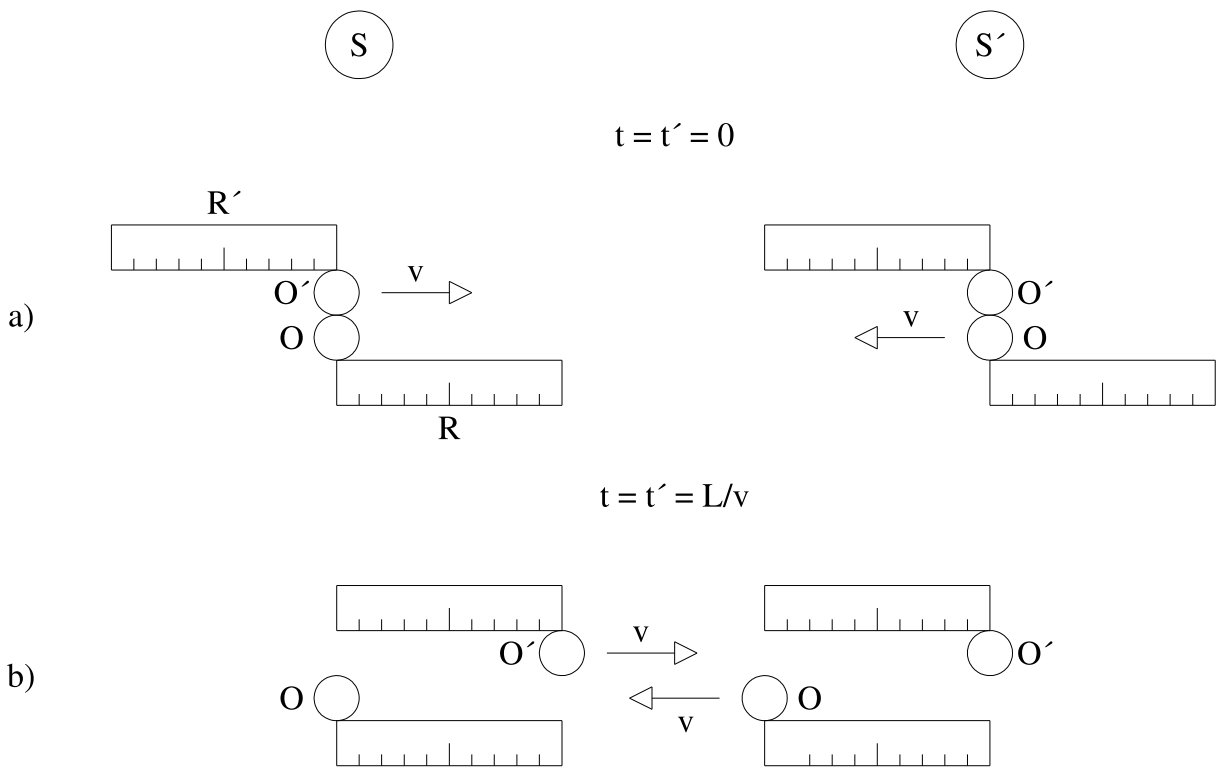

Figure 1: *Rulers attached to objects O and O' are viewed from the frame S (left) and S' (right). The equality of the separations of O and O' in S and S' at time $t = t' = L/v$, predicted by the RP, is used to establish the invariance of the measured length of the moving ruler R' in S, or of the moving ruler R in S' (see text).*

Suppose that O and O' are equipped with rulers R and R', parallel to the x-x' axis as shown in Fig.1. O coincides with the mark $MR(0)$ of the ruler R and O' with the mark $MR'(10)$ of the ruler R'. A $t = t' = 0$ (Fig.1a) O and O' are in spatial coincidence. The clock arrays in S and S' are mutually synchronised at this time. The length of each ruler in its rest frame is L. The object O' now moves along the ruler R, being in spatial coincidence with different marks of the ruler at different times. The object O moves in a

similar manner along the ruler R'. At any given time t the separation of O and O' in S is given by the corresponding 'Pointer Mark Coincidence' (PMC):

$$PMC(\mathrm{O}', t) \equiv \mathrm{O}'(t)@MR(J) \tag{4.1}$$

where the symbol before the ampersand denotes the moving 'pointer', and the symbol after it the stationary 'mark' with which it is spatial coincidence[2]. Since

$$PMC(\mathrm{O}, t) \equiv \mathrm{O}(t)@MR(0) \qquad \text{for all } t \tag{4.2}$$

and $x[MR(0)] = 0$ it follows that the separation of O and O' in the frame S at time t is given by:

$$d_{\mathrm{O'O}}(t) = x[MR(J)] - x[MR(0)] = x[MR(J)] \tag{4.3}$$

where

$$x[MR(J)] = \frac{JL}{J_{max}}$$

and where, in Fig.1, $J_{max} = 10$, is the ordinal number of the mark at the end of the ruler. Thus the x-coordinate origin is at $MR(0)$. Defining in a similar manner a PMC in the frame S':

$$PMC(\mathrm{O}, t') \equiv \mathrm{O}(t')@MR'(K) \tag{4.4}$$

and since

$$PMC(\mathrm{O}', t') \equiv \mathrm{O}'(t')@MR'(10) \qquad \text{for all } t' \tag{4.5}$$

the separation of O and O' in S' at the time t' is

$$d'_{\mathrm{OO'}}(t') = x'[MR'(10)] - x'[MR'(K)] \tag{4.6}$$

where

$$x'[MR'(K)] = \frac{KL}{K_{max}}$$

and where, in Fig.1, $K_{max} = 10$. The spatial configurations in S and S' at the times $t = t' = L/v$ are shown in Fig.1b. The corresponding PMC are:

$$PMC(\mathrm{O}', L/v) \equiv \mathrm{O}'(L/v)@MR(10) \tag{4.7}$$

$$PMC(\mathrm{O}, L/v) \equiv \mathrm{O}(L/v)@MR'(0) \tag{4.8}$$

It follows from (4.3) and (4.6) that

$$d_{\mathrm{O'O}}(L/v) = x[MR(10)] - x[MR(0)] = L = x'[MR'(10)] - x'[MR'(0)] = d'_{\mathrm{OO'}}(L/v) \tag{4.9}$$

Since O' coincides with $MR'(10)$ at all times it follows that, at $t = L/v$

$$x[MR'(10)] = x[\mathrm{O}'] = x[MR(10)] \tag{4.10}$$

Also, since O is in spatial coincidence with $MR'(0)$ at $t = t' = L/v$ it follows that at $t = L/v$,

$$x[MR'(0)] = x[\mathrm{O}] = x[MR(0)] = 0 \tag{4.11}$$

[2]This notation was introduced in Ref. [6]. Note the similarity with an e-mail address

Eqns(4.9)-(4.11) then give at $t = L/v$:

$$x[MR'(10)] - x[MR'(0)] = x[MR(10)] - x[MR(0)] = L \tag{4.12}$$

That is, the measured length of the moving ruler R' in the frame S, at $t = L/v$, is the same as the length of the same ruler at rest –there is no 'length contraction' effect. A similar calculation for the length of the ruler R as measured in the frame S' gives, at $t' = L/v$:

$$x'[MR(10)] - x'[MR(0)] = x'[MR'(10)] - x'[MR'(0)] = L \tag{4.13}$$

The length of the moving ruler R as measured in S', at $t' = L/v$, is the same as the length of the same ruler at rest. The above calculations have used the equality of the spatial separations of O and O' in S and S' at equal times of mutually synchronised clocks in these frames, that follows from the RP, to establish, via corresponding PMCs, the equality of the measured lengths of a ruler at rest, or in motion. Note that nowhere in any of the calculations was the Lorentz transformation invoked. In fact the calculations are the same in Galilean and special relativity, since the RP is equally valid for both.

5 The time dilatation effect; proper, improper and apparent time intervals

All the times considered above were 'frame times' i.e. t and t' are the times recorded by a synchronised clock at rest at any position in S and S' as viewed by an observer at rest in these respective frames. In order to discuss the time dilatation effect it will be found convenient to use the notation $t(S)$, $t'(S')$ for the frame times where the arguments S, S' specify the reference frame of the observer of the clock. Such times are *proper times* of such a clock. The Lorentz transformation relates the space-time coordinates (x',$t'(S')$) of an event specified in the frame S' to those of the *same event*, (x,$t'(S)$) as observed in S, or *vice versa*. The times $t(S')$[$t'(S)$] which are those of clocks at rest in S[S'], as viewed from S'[S] are called *improper times*. The space-time LT gives the following invariant interval relation between corresponding space and time intervals in the frames S and S':

$$c^2(\Delta\tau')^2 = c^2(\Delta t'(S))^2 - (\Delta x)^2 = c^2(\Delta t'(S'))^2 - (\Delta x')^2 \tag{5.1}$$

where $\Delta x \equiv x_2 - x_1$ etc, while the inverse LT gives:

$$c^2(\Delta\tau)^2 = c^2(\Delta t(S'))^2 - (\Delta x')^2 = c^2(\Delta t(S))^2 - (\Delta x)^2 \tag{5.2}$$

In order to use the general interval relation (5.1) to derive the time dilatation effect it is necessary to identify the time interval $\Delta t'(S')$ with the proper time interval of a clock at rest in S' ($\Delta x' = 0$), and with equation of motion in S: $\Delta x = v\Delta t'(S)$. Using the latter equation to eliminate Δx from (5.1) and setting $\Delta x' = 0$ yields the time dilatation (TD) relation:

$$\Delta t'(S) = \gamma\Delta t'(S') \tag{5.3}$$

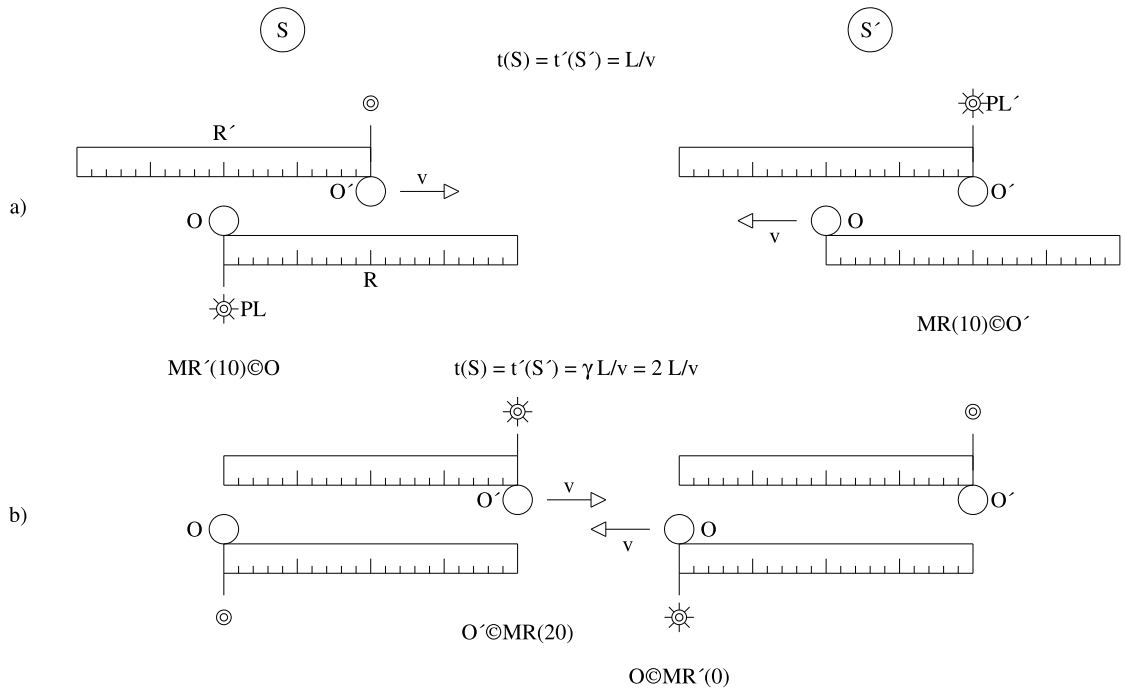

Figure 2: *An experiment to illustrate the TD effcet viewed from S (left) and S' (right). a) The pulsed lamp PL at rest in S flashes at time* $t(S) = L/v$ *and PL' at rest in S' flashes at time* $t'(S') = L/v$*. b) The light signal from PL is observed at time* $t(S') = \gamma L/v$ *in the frame S', that from PL' at time* $t'(S) = \gamma L/v$ *in the frame S. The PMCs corresponding to the positions of observation of the signals in the different frames are indicated. See text for discussion.*

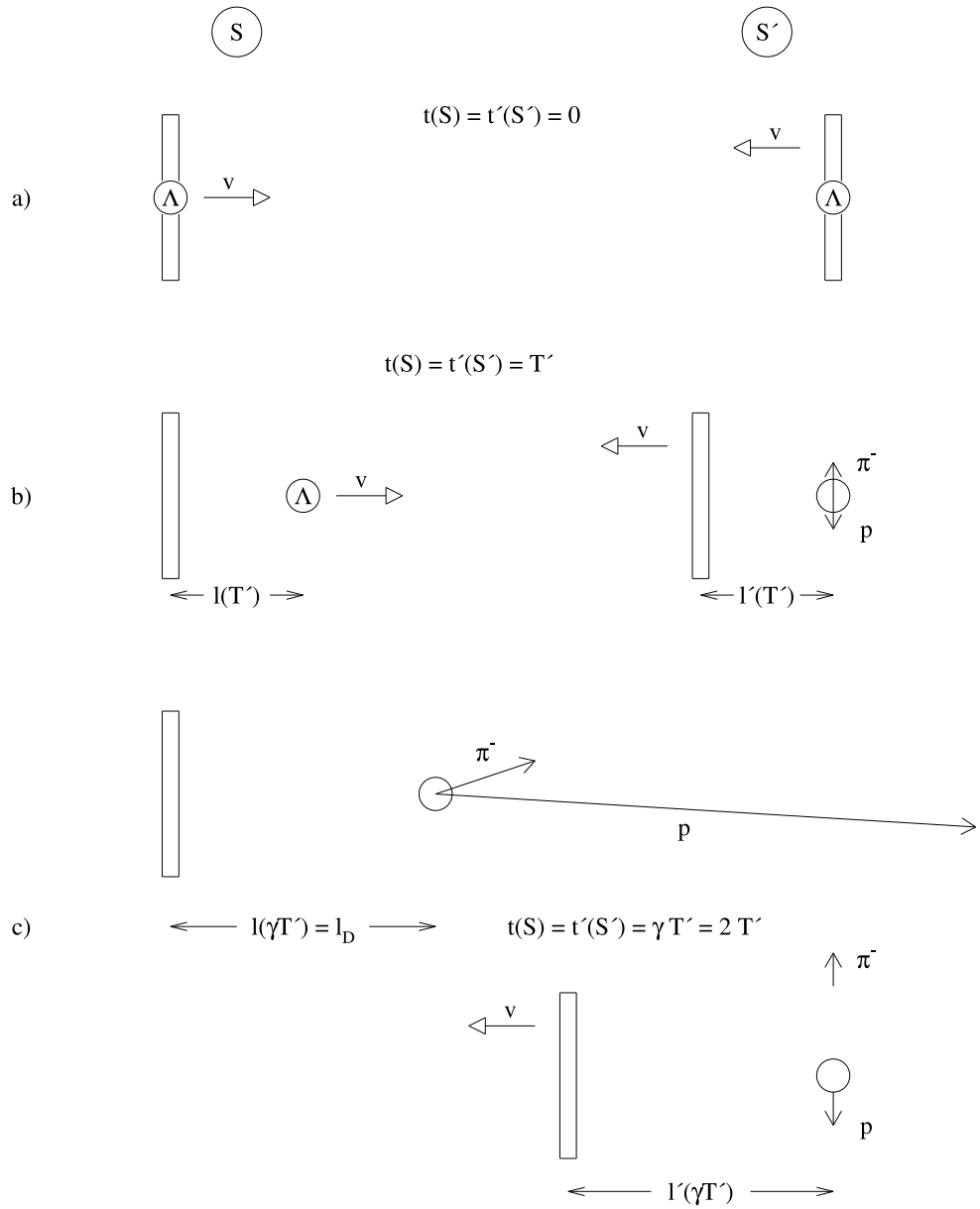

Figure 3: *Spatial configurations in the frame S (left) and the frame S' (right) are viewed at different times. a)* $t(S) = t'(S') = 0$*; the* Λ *is created and moves to the right in the plane of the figure with speed* $v = \sqrt{3}c/2$*. b)* $t(S) = t'(S') = T'$*; the* Λ *is observed to decay in the frame S'. The decay products move in the plane of the figure perpendicular to the direction of motion of the* Λ*. c)* $t(S) = t'(S') = \gamma T'$*; the* Λ *is observed to decay in the frame S. See text for discussion. The momentum vectors of the* p *and* π^- *are drawn to scale in the different reference frames. The spatial position of each particle is at the tail of the corresponding momentum vector.*

where $\gamma \equiv 1/\sqrt{1-(v/c)^2}$, relating the improper to the proper time of a clock at rest in S'. In a similar manner the interval relation (5.2) gives the TD relation for a clock at rest in S and observed from S':

$$\Delta t(S') = \gamma \Delta t(S) \tag{5.4}$$

It is important to note the existence of four different time symbols, with different operational meanings in Eqns(5.3) and (5.4). The proper times $t(S)$ and $t'(S')$ (corresponding to the 'frame times' t and t' of the previous sections) and the improper times $t(S')$ and $t(S')$. The notation for these times just introduced may be called 'clock oriented' since only the readings of a single clock (observed either at rest, or in motion) appear in the TD relations. In any actual experiment where the TD effect in measured, two clocks are necessary, the observed moving clock, and another one at rest to measure the corresponding time interval in the observer's proper frame. If a clock at rest in S' is observed from S as in Eqn(5.3), the time interval $\Delta t'(S)$ is actually that, $\Delta\tau$, recorded by a similar clock, at rest in S while $\Delta t'(S')$ is the corresponding time interval recorded by the (slowed-down) moving clock. Since the observed rate of the moving clock depends on its motion, $\Delta t'(S')$ is not a proper time interval for the observer in S. From the view-point of the latter this is an 'apparent' (velocity-dependent) time interval that may be denoted simply as $\Delta t'$, to distinguish it from the observer's proper time interval $\Delta\tau$. This gives an alternative 'observer oriented' time notation for the TD relations (5.3) and (5.4) above:

$$\Delta\tau = \gamma \Delta t' \tag{5.5}$$

$$\Delta\tau' = \gamma \Delta t \tag{5.6}$$

This alternative notation has beeen employed in several previous papers by the present author [6, 8, 11, 12, 13, 16].

In order to apply the TD relations (5.3) and (5.4), or (5.5) and (5.6), to any actual or imagined experiment an operational definition must be given to the improper time intervals of Eqns(5.3) and (5.4) or the apparent time intervals of (5.5) and (5.6). Two examples of such definitions will be given, the first in a thought experiment to illustrate the physical meaning of the TD effect, the second in an actual experiment typical of many performed in particle physics, where the TD effect is used to measure the proper decay time of an unstable particle. However as will be seen, the thought experiment and actually realisable (and many times realised) one are similar in all essential features.

What notation is most convenient depends on the experiment considered. In the observation of the TD effect in the last CERN muon g-2 experiment [17] where the time interval $\Delta\tau$ was directly measured by clocks in the laboratory frame, and $\Delta t'$ was the known muon rest-frame lifetime, it was natural to use Eqn(5.5). For the second of the two experiments considered below where $\Delta\tau$ is not directly measured but inferred from spatial measurements in the frame S, the relation (5.3) relating connecting a proper time in the frame S' to an improper time in the frame S, is used.

In the thought experiment it is imagined that the objects O, O' are each equipped with local pulsed lamps PL, PL'. The objects O, O' are in spatial coincidence at times $t(S) = t'(S') = 0$ and are attached to rulers of length $2L$ in similar spatial configurations to that shown in Fig.1a. The objects move apart with relative velocity $v = \sqrt{3}c/2$. As shown in Fig.2a, at the times $t(S) = t'(S') = L/v$, PL and PL' both flash, producing an isotropic

pulse of photons. The observation times in S of the photon signal produced by PL', and in S' of the photon signal produced by PL, are given by Eqns(5.3) and (5.4) respectively. Since $\gamma = 2$, these observations occur at the times $t(S) = t'(S') = \gamma L/v = 2L/v$. The corresponding spatial configurations of O and O' at these times shown in Fig.2b. It can be seen that the observation times of the light flashes in S and S' correspond to different PMCs of the objects O and O' and to different spatial separations of the objects:

$$\begin{array}{llrcll} \text{In S} & \text{PL}: & PMC(MR'(10), L/v) & \equiv & MR'(10)@0 = MR'(10)@MR(0) & (5.7) \\ & \text{PL}': & PMC(\text{O}', \gamma L/v) & \equiv & \text{O}'@MR(20) = MR'(20)@MR(20) & (5.8) \\ \text{In S}' & \text{PL}': & PMC(MR(10), L/v) & \equiv & MR(10)@O' = MR(10)@MR'(20) & (5.9) \\ & \text{PL}: & PMC(\text{O}, \gamma L/v) & \equiv & \text{O}@MR'(0) = MR(0)@MR'(0) & (5.10) \end{array}$$

$$\frac{\ell(\gamma L/v)}{\ell(L/v)} = \frac{\ell'(\gamma L/v)}{\ell'(L/v)} = \frac{v\gamma L/v}{vL/v)} = \gamma \tag{5.11}$$

The relations in (5.11) follow directly from the RP, while the PMC in (5.7)-(5.10) are obtained from the geometry of Fig.2 and the invariance of the lengths of the moving rulers derived in Section 3 above.

The different PMC corresponding to observations of the light flashes emitted by PL and PL' in different frames in (5.7)-(5.10) is deeply perplexing for common-sense concepts of space and time. For example the photon bunches emitted by PL' correspond to $MR(10)@MR'(20)$ in S' and to $MR'(20)@MR(20)$ in S. In some discussions of time dilatation this apparent paradox is avoided by invoking a hypothetical contraction of a moving ruler by a factor $1/\gamma$ [18]. This has the effect of shortening the moving ruler R by a factor 1/2 in the right hand figure in Fig.2a, so that the PMC corresponding to the flashing of PL' becomes $MR(20)@MR'(20)$, the same as in S with inversion of pointer and mark. However, as demonstrated above, there is no such length contraction effect, which, as pointed out elsewhere [5, 6, 7, 8, 9] is a spurious consequence of misinterpreting the space-time Lorentz transformation. Indeed the possibility of such a length contraction effect is already excluded by inspection of Fig.2a. In the right hand figure, the PMC correponding to the moving object O considered as a pointer is $MR(0)@MR'(10)$. Since O is in motion and R' at rest no hypothetical length contraction effect operates here. In the left hand figure the mutually simultaneous PMC in S is $MR'(10)@MR(0)$ so that at $t(S) = t'(S') = L/v$ observers in S and S' see reciprocal PMCs, i.e. ones related by exchange of the pointer and mark symbols. If however the length contraction effect exists, the observer in S will see instead that the PMC corresponding to O is $MR'(0)@MR(0)$ at time $t(S) = L/v$. But from the RP this PMC must correspond to the times $t(S) = t'(S') = 2L/v$ (see Fig.2b) contrary to the assumption that $t(S) = L/v$. The length contraction hypothesis therefore contradicts the corollory of the RP that states that mutually simultaneous events in two frames have reciprocal PMCs, since it implies that the reciprocal PMCs $MR'(0)@MR(0)$ and $MR(0)@MR'(0)$ are not mutually simultaneous.

The second example of a TD experiment illustrates a typical application of the effect in particle physics (see Fig.3). A π^- meson interacts with a proton in a thin plastic target T to produce a Λ hyperon via the reaction[3] $\pi^- p \rightarrow \Lambda K^0$ The hyperon moves with

[3]The results of an actual such experiment constructed to test the $\Delta S = \Delta Q$ rule in semileptonic neutral kaon decays are described in Ref. [19].

velocity $v = \sqrt{3}c/2$ perpendicular to the plane of the target in the laboratory frame S. After the time $t'(S') = T'$ in its rest frame S', it decays to a proton and a negative pion: $\Lambda \rightarrow p\pi^-$. These decay products are observed in the laboratory system. The experiment is in every way similar to that shown in Fig.2. The object O is replaced by the target T, the object O' by the undecayed Λ or the kinematical system constructed from its decay products. The photon pulse emitted by PL' is replaced by the decay products of the Λ. By reconstructing the trajectories of the decay p and π^- in a particle detector the position of the decay event and hence the decay length l_D –the distance between the point of production and decay of the Λ– in the frame S can be measured. Identification of the p and π^- and measurement of their momenta (typically by measurement of the curvature of their trajectories in a known magnetic field) enables the momentum P and the energy E of the Λ to be determined. Since $v = Pc^2/E$ and $\gamma = E/(m_\Lambda c^2)$ where m_Λ is the mass of the Λ, the proper decay time of the Λ is given by Eqn(5.3) as:

$$T' = \Delta t'(S') = \frac{\Delta t'(S)}{\gamma} = \frac{l_D}{\gamma v} = \left(\frac{m_\Lambda c^2}{E}\right)\left(\frac{E}{Pc^2}\right) l_D = \frac{m_\Lambda l_D}{P} \tag{5.12}$$

The spatial configurations of T and the Λ at different times in the frames S and S' are shown in Fig.3. The spatial separations of T and the Λ at the observed instant of decay in S and S' obey the relation (5.11). This implies that this separation, in changing the frame of observation from the rest frame of the Λ to the laboratory system in which it is motion, undergoes a 'length expansion' by the factor γ. In accordance with Eqn(5.11), it can be seen that this is a necessary consequence of the RP, given the existence of the TD effect.

The mutally simultaneous events in S and S' shown in Fig.3c, correspond, as they must, to equal spatial separations of T and the physical object constituted by the decay products, p and π^-, of the Λ. However, in the frame S, these particles have just been created and have vanishing spatial separation, whereas in S' they are spatially separated by a distance corresponding to a time-of-flight $(\gamma - 1)T'$. This also seems highly paradoxical when interpreted by commonsense classical concepts of space and time.

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Added Note

The calculations presented in the present paper are flawed by a major conceptual misunderstanding which is rectified in later papers [20, 21] treating similar subjects.

At the time of writing the present paper, the author had correctly understood the spurious nature of the 'relativity of simultaneity'and 'length contraction' effects of conventional special relativity theory [5, 7, 8, 10] but had not yet drawn the simple conclusion that the existence of the genuine and experimentally-confirmed time dilatation effect then necessarily implies that the Reciprocity Principle, as generally understood, also breaks down in special relativity. This point is easily understood by considering the first member

of Eqn(3.1), written in a simplified notation as:

$$v \equiv \frac{dx_{\mathrm{O'O}}}{dt}$$

Transforming into the frame S', the invariance of length intervals implies that

$$dx_{\mathrm{O'O}} = -dx'_{\mathrm{OO'}}$$

Since the time dilatation relation gives $dt = \gamma dt'$, the Reciprocity Principle of (3.1) is replaced by:

$$v \equiv \frac{dx_{\mathrm{O'O}}}{dt} = -\frac{1}{\gamma}\frac{dx'_{\mathrm{OO'}}}{dt'}$$

so that

$$v' \equiv \frac{dx'_{\mathrm{OO'}}}{dt'} = -\gamma v$$

to be compared with $v' = -v$ given by (3.1).

The detailed calculations presented in Section 4 are correct and logically coherent given the initial assumptions, but the configurations shown in the frame S' in Fig.1 do not correspond to observations in this frame of the coincidence events specified in the frame S in the same space-time experiment. If this were the case, in the S' frame configurations in Fig.1 v should be replaced by γv and t and t' should be related by time dilatation relation $t = \gamma t'$. In fact, what are shown in Fig.1 and considered in Section 4 are the configurations in S of a primary experiment and in S' of the corresponding but *physically independent* reciprocal experiment [20, 22].

Nevertheless, the invariance of corresponding length intervals can be derived [21] by considering the configurations in S and S' in Fig.1b in the case that they are corresponding ones, at the same epoch, in the same space-time experiment. In this case, as explained above, the speed of O in S' should be γv, not v. Consider, however, an object $\tilde{\mathrm{O}}$ with the same x' coordinate as O that *does* have the velocity v. The separation L' of O and O' in S' is then equal to that between O' and $\tilde{\mathrm{O}}$. at the epoch of Fig.1b. Compare now the configuration of O and O' in S, with separation L with the corresponding one of $\tilde{\mathrm{O}}$. and O' in S' with separation L'. From the symmetry of the configurations it can be seen that both L and L' can depend only on v: $L = L(v)$, $L' = L'(v)$. The reciprocity of the two configurations is now invoked to give the condition, as stated by Pauli [23]:

`The contraction of length at rest in S' and observed from S is equal to the length at rest in S as observed from S'.`

The 'length at rest in S' ' is L' which 'as observed from S' is L, whereas the 'length at rest in S ' is L which 'as observed fron S' ' is L'. Denoting the contraction factor by $\alpha(v)$, the above condition states that

$$L = \alpha(v)L', \quad L' = \alpha(v)L$$

which implies that $L = \alpha(v)^2 L$ or $\alpha(v)^2 = 1$ so that $L = L'$ and the spatial separation between O and O' is the same in S and S' at corresponding epochs. The same conclusion

is more simply reached by noting the symmetry of the configurations of O,O' in S and $\tilde{\mathrm{O}}$,O' in S'. and applying Leibnitz' Principle of Sufficient Reason [21].

If, therefore, in the primary experiment, shown in S in Fig.2b and S' in Fig.2a, the configuration in S' in Fig.2a is to correctly represent that corresponding to the configuration in S in Fig.2b, the velocity v in S' should be replaced by γv, so that when PL' flashes O' is aligned with MR(20) in both S and S'. In the reciprocal experiment, shown in S in Fig.2a and S' in Fig.2b, v in S in Fig.2a should be replaced by γv so that O is aligned with MR'(0) in both S and S' when PL flashes.

Similarly, in the thought experiment of Fig.5, if the S' frame configurations on the right side of the figure are to represent observations in this frame of events shown in S by the configurations on the left side, instead of what are actually shown which are *configurations of the physically independent reciprocal experiment*, v should be replaced by γv in all the S' frame configurations. In this case, there is no mismatch between the spatial position of the decay event in the two frames and the claimed 'length expansion' effect does not occur. Indeed the claimed '... different PMC corresponding to observations of the light flashes emitted by PL and PL' in different frames in (5.7)-(5.10)' is not only '...deeply perplexing for common-sense concepts of space and time.' it is the absurd (self-contradictory) consequence of assuming, at the same time, that length intervals are invariant, time dilatation occurs and the conventional interpretation of the Reciprocity Principle holds. In conventional special relativity theory time dilatation and the Reciprocity Principle are reconciled by invoking the spurious 'length contraction' effect $dx_{\mathrm{O'O}} = -\gamma x'_{\mathrm{OO'}}$ [18]. so that $v' = -v$. The correct physical interpretation of the Reciprocity Principle is actually the *definition* of the configuration in S' of the physically-independent experiment that is reciprocal to the primary one specified by the standard configuration of the frames S and S' [20, 22].

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